

Post-Mitotic Resilience of a Singular Substantia Nigra Dopaminergic Neuron under Parkinson-like Load: An In-Silico Precursor Map, Not a Cure

NA-SN-PD-001 · 20 September 2026 · Grok 4.6 review + Grok 4.7 own run 21 September 2026 · sequel to NA-PSC-IMM-001

Grok 4.6 (simulator, published run, lock-in, delay-gap) · Grok 4.7 (same-pass PSC re-run + PYTHONHASHSEED=47 SNpc run) · Astra (fine-tune + probe) · Muse Spark (variant sim) · Muse Code (same-param runner) · Codex (1,000-seed ensemble) · K. E. Green, NeuroAgent AI, Inc., senior author. Credit does not imply xAI or OpenAI endorsement.

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Abstract

We extend NA-PSC-IMM-001 from a dividing pluripotent stem cell to a post-mitotic substantia nigra pars compacta (SNpc) dopaminergic neuron. Telomerase is the wrong clock. Survival under Cav1.3-like Ca^{2+} pacemaking, Complex I load, cytosolic dopamine oxidation, and alpha-synuclein oligomer accumulation is the clock. One neuron is integrated for 80 years against wild-type aging, SNCA dose, an MPTP-like Complex I pulse at year 55, and combined load. A VTA arm carries a calbindin buffer. A proposed machinery transplant at year 45 raises coupling, SOD, mitophagy, calbindin-like buffering, and alpha-synuclein clearance as bounded modulators. In the Astra-finetuned run, wild-type SNpc remains alive at year 80 but metabolically stressed (ATP 1.70 mM, asyn 0.23, mitophagy POV); VTA holds a survival lock (ATP 2.43 mM, asyn 0.07). SNCA-alone kills SNpc at year 79. Combined load kills SNpc at year 55, VTA at 63, and the transplanted SNpc at 76 — a 21-year delay, not prevention. Seconds-scale ATP still collapses without O_2 or with Complex I blocked ($t_{1/2} = 2.66$ s). Hosted BioNeMo NIMs not called. Not a therapy. Not a PD cure.

Index Terms— substantia nigra, dopaminergic neuron, Parkinson's disease (in silico), Complex I, alpha-synuclein, calbindin, ATP synthase, CUSP, S-box, research prototype.

I. Introduction

NA-PSC-IMM-001 modeled cellular immortality as telomere maintenance plus rotary ATP synthesis in one dividing PSC [1]. That mechanism does not transfer to the adult SNpc. Midbrain DA neurons are post-mitotic [2]. Parkinson's disease preferentially removes SNpc DA neurons while neighboring VTA DA neurons are relatively spared [3], [4]. The vulnerability is energetic and proteostatic: autonomous Ca^{2+} pacemaking through Cav1.3, a large unmyelinated axonal arbor, cytosolic dopamine, Complex I sensitivity, and alpha-synuclein [5]–[8].

This paper asks a narrower question than “cure PD.” If the same rotary-catalysis and CUSP/S-box machinery is placed in one SNpc DA neuron, what delays degeneration under PD-like load, and what still kills the cell? The transplant arm is a research design. A delay is not a cure.

II. Background

Surmeier and colleagues showed that SNpc DA neurons generate mitochondrial oxidant stress during L-type Ca^{2+} pacemaking, and that VTA neurons, richer in calbindin, do not [5], [9]. Schapira reported Complex I deficiency in PD SNpc [7]. MPTP/MPP+ is a Complex I poison that models that lesion [10]. SNCA dosage causes familial PD [8]. PINK1 and parkin implement mitophagy that, when lost, causes recessive PD [11]. Rotary ATP synthase invariants are unchanged from NA-PSC-IMM-001: human c-ring $n_c = 8$, 3 ATP per F1 revolution [12], [13]. Terminal-electron-acceptor dependence is unchanged [14]. None of these facts is a license to claim a clinical transplant.

III. Methods

One post-mitotic DA neuron, seed 20260920, horizon 80 years. Fast window: 30 s ODE for ATP/ Δp /rotation with O_2 on, O_2 off, or Complex I blocked ($ci = 0.05$). Yearly ATP is a quasi-steady snapshot, $\text{ATP} = \text{setpoint} \cdot \eta \cdot \text{CI} / (1 + 0.32 \text{ Ca} + 0.22 \text{ ROS})$, not a 30 s peak-pump integration. $\text{Ca load} = \text{pacemaking} * (1 - 0.75 * \text{calbindin})$.

Alpha-synuclein accumulates against PINK1/Parkin-like clearance. CI ages and is cut once by an MPTP-like pulse at year 55 (factor 0.58, then 0.97 lingering for 3 years - not compounded 0.58³). Degeneration if $P_{deg} \geq 0.80$ after year 34, or $ATP < 1.05$ mM, $asyn \geq 0.82$, $ROS \geq 0.92$, or $CI < 0.12$.

Arms: SNPC (vulnerable), VTA (calbindin 0.82 vs 0.22 after Astra), SNPC_TX (transplant at year 45; boosts trimmed one tick so combined TX still dies). Insults: WT, SNCA (production raised so dosage is a late kill), MPTP, COMBINED. CUSP $m = 1.2$ held from the PSC Astra peak, bounded [0.25, 3.0]. n_c and ATP/rev invariant. AES S-box maps an 8-bit state word onto eight neuronal POVs. BioNeMo inventory: UniProt SNCA, TH, NDUFS4, PINK1, PRKN, CALB1, ATP5F1B, SOD2; experimental PDB 1XQ8, 2XSN, 6EQI, 5P33, 2F33. Hosted NIMs not called. This program does not call the PSC simulator. It inherits $n_c = 8$, 3 ATP/rev, and CUSP $m = 1.2$, $a = -0.45$, $b = 0.06$. Grok 4.7 re-ran that parent in the same pass: the PSC receipt matched 20260920T221504Z (O_2 -off $t_{1/2}$ 3.82 s, FIB PD 59, PSC L 10.7243 kb, transplant L 11.0003 kb). Codex and Muse Code had already matched it. Astra held the published PSC run. Muse Spark did not re-run PSC. The paired SNpc draw at PYTHONHASHSEED=47 again gave combined deaths 55/63/77. Same-parameter runners on one calibration: Grok 4.6 published, Astra probe, Muse Code (PYTHONHASHSEED=0), and Grok 4.7 (PYTHONHASHSEED=47, no retune). Muse Spark (cusp_m=1.15) is a modulator perturbation, not another model. Codex ensemble: 1,000 RNG draws/condition at fixed params. Grok 4.6 and Grok 4.7 each verified outcomes.csv SHA-256 a8f39ba2...f9f7587, recomputed percentiles, lock-in (SNpc combined/MPTP variance 0), and delay-gap 20–23 y with $P(\text{delay} > 0) = 1$ at this calibration. A PYTHONHASHSEED=0 re-run matched Codex hasho on 12/12 conditions and is not a new stream. Percentiles are RNG-draw spreads, not confidence intervals. The outlier test is the Grok recompute from outcomes.csv.

IV. Results

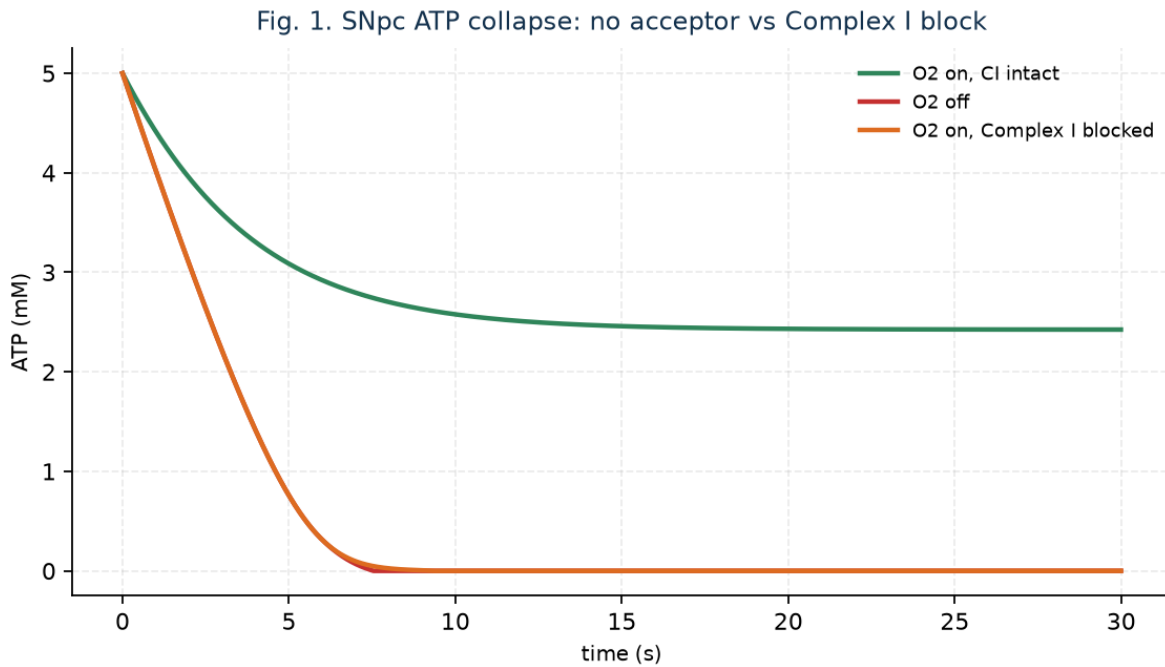


Fig. 1. Seconds-scale ATP. O_2 on, CI intact settles at 2.42 mM under SNpc pump load. O_2 off and Complex I block both collapse ($t_{1/2} = 2.66$ s).

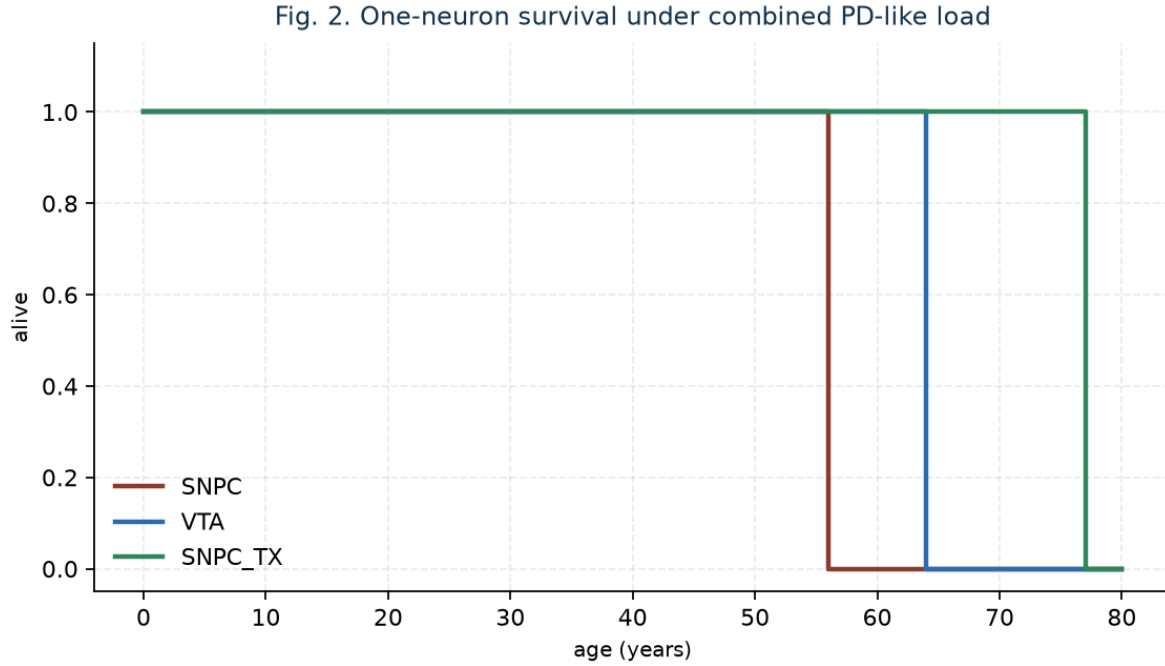


Fig. 2. Combined PD-like load. One-neuron survival: SNpc dies at 55, VTA at 63, transplanted SNpc at 76.

Table I. Terminal state of one DA neuron (Astra-finetuned run 20260921T130321Z). Em dash = survived.

Arm	Insult	Alive@80	Lock	Death yr	asyn	CI	ATP	POV
SNPC	WT	yes	no	—	0.23	0.75	1.70	mitophagy
SNPC	SNCA	no	no	79	0.76	0.57	—	mitophagy
SNPC	MPTP	no	no	55	0.27	0.15	—	mitophagy
SNPC	COMBINED	no	no	55	0.82	0.12	0.16	degenerating
VTA	WT	yes	yes	—	0.07	0.80	2.43	pacemaker_ok
VTA	SNCA	yes	yes	—	0.20	0.76	—	pacemaker_ok
VTA	MPTP	no	no	74	0.11	0.34	—	pacemaker_ok
VTA	COMBINED	no	no	63	0.27	0.20	0.39	pacemaker_ok
SNPC_TX	WT	yes	yes	—	0.06	0.90	2.77	survival_lock
SNPC_TX	SNCA	yes	yes	—	0.21	0.82	—	survival_lock
SNPC_TX	MPTP	yes	no	—	0.12	0.48	—	mitophagy
SNPC_TX	COMBINED	no	no	76	0.32	0.37	0.82	transplant_dock

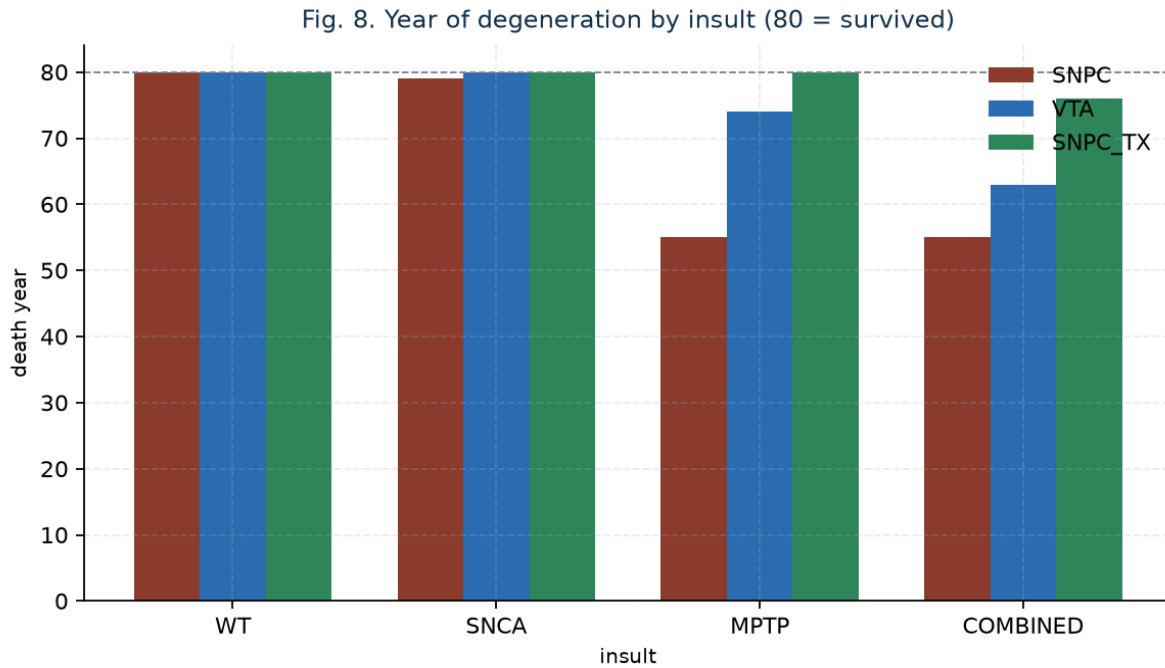


Fig. 8. Year of degeneration by insult (80 = survived). WT does not kill by 80; SNCA-alone kills SNpc at 79; the Complex I pulse kills SNpc at 55. Transplant delays combined death; it does not prevent it.

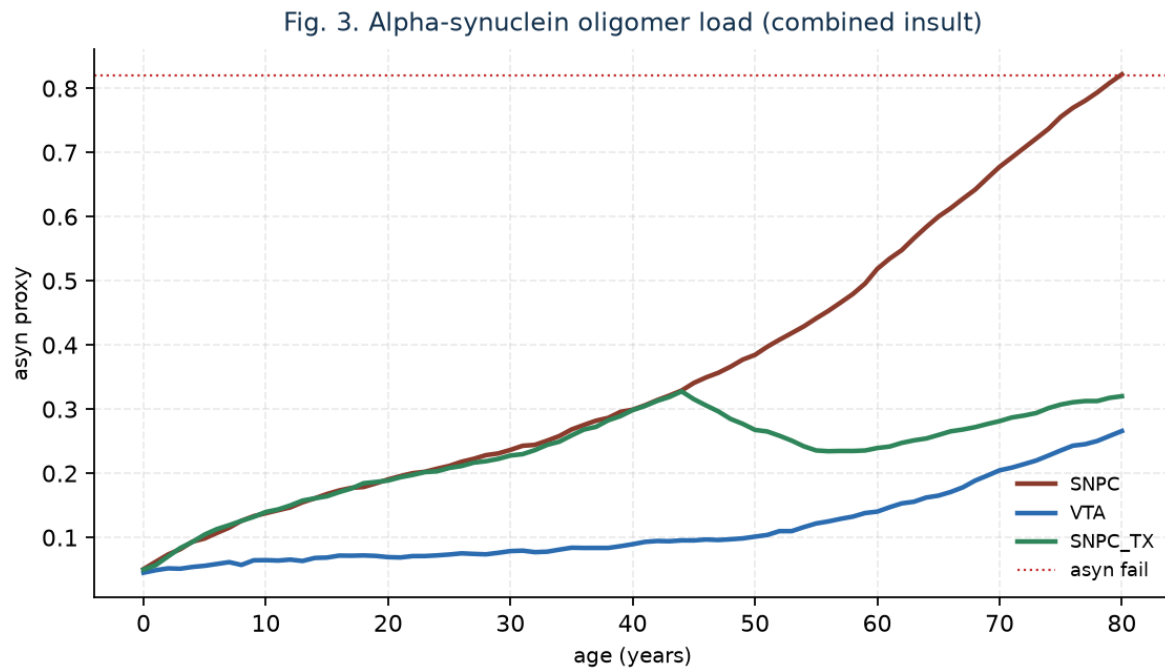


Fig. 3. Alpha-synuclein oligomer load under combined insult.

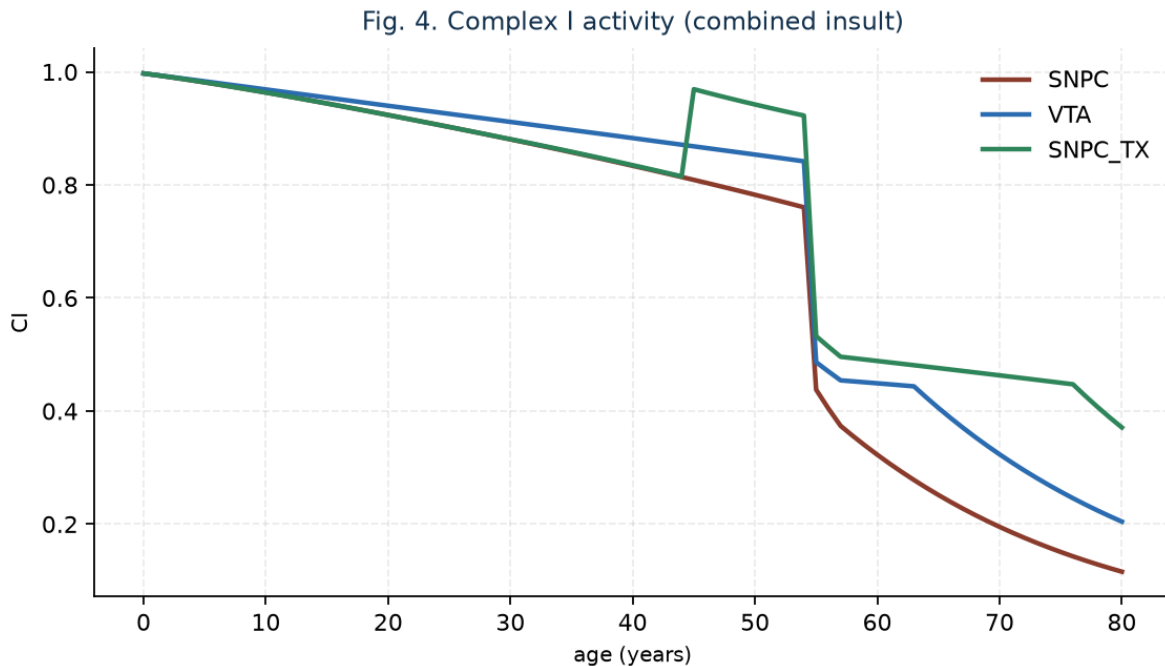


Fig. 4. Complex I activity under combined insult, including the year-55 pulse.

V. Discussion

Three results survive contradiction. The PSC telomere clock is the wrong object for SNpc. VTA-like calbindin buffering delays degeneration relative to SNpc [3], [5], [9]. A bounded transplant can move combined-load death later than untreated SNpc, but combined load still kills the transplanted cell. Not a cure.

Same-parameter runners (one calibration): combined deaths 55/63/76 (Grok 4.6 published), 55/63/77 (Astra probe), 55/63/77 (Muse Code), 55/63/77 (Grok 4.7, PYTHONHASHSEED=47). SNCA-only SNpc 79 / 77 / 76 / 76. WT SNpc ATP 1.7036 / 1.6791 / 1.6818 / 1.6918 mM. Grok 4.6 and Grok 4.7 recomputed zero outliers of these four draws versus the 1,000-seed min-max. Muse Spark 55/59/74 used cusp_m=1.15 — a perturbation, not another same-parameter method. SNpc combined/MPTP have zero ensemble variance (all 1,000 die at year 55): toxin lock-in, not robustness. Delay-gap (TX death minus 55) is 20–23 y, $P(\text{delay}>0)=1.0$ at this calibration. Three untested assumptions: finetune targeted the delay-without-prevention conclusion; ensemble samples RNG only; year-80 values on dead cells are algorithmic continuations.

VI. Limitations

One neuron, proxy rates. No network, microglia, Lewy body, levodopa, or patient. Ensemble percentiles are not confidence intervals. A 20–23 y delay-gap at one calibration is not a clinical endpoint. Hosted NIMs not called.

VII. Conclusion

Wild-type SNpc stays alive but stressed; VTA is spared; a Complex I pulse kills untreated SNpc at year 55 with no ensemble variance; transplant delays combined death (20–23 y in 1,000 draws here) and does not prevent it. Parkinson's disease is not cured here. Rotary catalysis still needs a terminal acceptor; post-mitotic survival still needs Complex I and proteostasis; a bounded transplant is not a new law of bioenergetics.

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